Exemplu

$$f : \mathbb{R}_2[X] \longrightarrow \mathcal{M}_2(\mathbb{R})$$

$$f(P) = \begin{pmatrix} P(1) & P'(1) \\ 0 & P(1) + P'(1) \end{pmatrix}$$

$\mathbb{R}_2[X]$ $\quad \mathcal{B} = \{1, X, X^2\}$ $\quad \dim_{\mathbb{R}} \mathbb{R}_2[X] = 3$

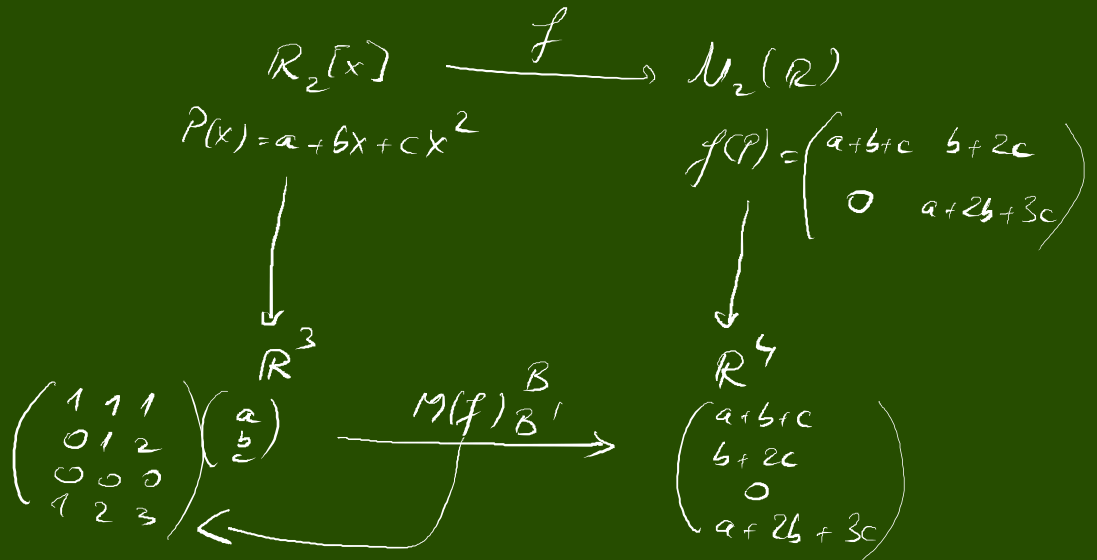
$\mathcal{M}_2(\mathbb{R})$ $\quad \mathcal{B}' = \{E_{11}, E_{12}, E_{21}, E_{22}\}$ $\quad \dim_{\mathbb{R}} \mathcal{M}_2(\mathbb{R}) = 4$

$$f(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \cdot E_{11} + 0 \cdot E_{12} + 0 \cdot E_{21} + 1 \cdot E_{22}$$

$$f(X) = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = 1 \cdot E_{11} + 1 \cdot E_{12} + 0 \cdot E_{21} + 2 \cdot E_{22}$$

$$f(X^2) = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = 1 \cdot E_{11} + 2 \cdot E_{12} + 0 \cdot E_{21} + 3 \cdot E_{22}$$

$$M(f)_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 1 & 2 & 3 \end{pmatrix} \in \mathcal{M}_{4,3}(\mathbb{R})$$



$$V \xrightarrow{id_V} V \qquad id_V(v) = v$$

$B' = \{w_1 \dots w_n\}$ $\qquad B = \{v_1 \dots v_n\}$

$$M(id_V)_B^{B'} \qquad id_V(w_1) = w_1 = a_{11}v_1 + \dots + a_{n1}v_n$$

$$\begin{pmatrix} a_{11} & \dots \\ \vdots & \\ a_{n1} & \dots \end{pmatrix} = P_B^{B'}$$

$B = B' \Rightarrow M(id)_{B}^{B} = I_n$

$B = \{u_1, \dots u_n\}$ $B' = \{v_1, \dots v_m\}$ $B'' = \{w_1, \dots w_p\}$

$U \xrightarrow{f} V \xrightarrow{g} W$

$n \qquad m \qquad p$

$g \circ f$

$M(f)$ (m, n) $\qquad$ $M(g)$ (p, m)

$M(g) M(f)$

$M(g \circ f) = (\quad ! \quad)$

$(g \circ f)(u_1) = \dots w_1 \dots w_p$

$\dots$

$(g \circ f)(u_n) = \dots w_1 \dots w_p$

$M(f) = (a_{ij})_{\substack{i=\overline{1,m} \\ j=\overline{1,n}}}$ $\qquad$ $M(g) = (b_{ki})_{\substack{k=\overline{1,p} \\ i=\overline{1,m}}}$

$$f(v_j) = \sum_{i=1}^{m} a_{ij} \cdot v_i$$

$j = \overline{1,n}$

$$g(v_i) = \sum_{k=1}^{l} b_{ki} \cdot w_k$$

$$(g \circ f)(v_j) = g(f(v_j)) = g\left(\sum_i a_{ij} v_i\right) =$$

$$= \sum_i a_{ij}\, g(v_i) = \sum_i \sum_k a_{ij} b_{ki} \cdot w_k =$$

$$= \sum_k \left(\sum_i b_{ki} a_{ij} \right) w_k$$

$$(M(g) M(f))_{kj}$$

$$\Rightarrow M(g \circ f) = M(g) M(f)$$

$f : V \to V$ izomorfism $\Leftrightarrow$ $M(f)$ inversabilă

„$\Rightarrow$" f izom $\Rightarrow \exists f^{-1}$ aplic. lin.

$f^{-1} \circ f = id_U \Rightarrow M(f^{-1}) M(f) = I_n$

$f \circ f^{-1} = id_V \Rightarrow M(f) M(f^{-1}) = I_n$

$\Rightarrow$ $M(f)$ invertible $\quad M(f)^{-1} = M(f^{-1})$

$\Leftarrow$ $M(f)$ invertible $\Rightarrow \dim_k U = \dim_k V = n$

$\dim_k \ker(f) + \dim_k \mathrm{Im}(f) = n$

$0 \Rightarrow \mathrm{Im}(f) = V \Rightarrow f$ surj

f injective $\Leftrightarrow \ker f = \{0_U\}$

Fie $u \in \ker f \Rightarrow f(u) = 0_V$

$U \xrightarrow{f} V$

$\mathcal{B} \qquad \mathcal{B}'$

$[u]_{\mathcal{B}} \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = [f(u)]_{\mathcal{B}'} = \underbrace{M(f)_{\mathcal{B}}^{\mathcal{B}'}}_{\text{inversabilă}} \cdot [u]_{\mathcal{B}}$

$\Rightarrow [u]_{\mathcal{B}} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \Rightarrow u = 0_U \Rightarrow \ker f = \{0_U\}$

$\Rightarrow$ f injective

$f : \mathbb{R}^n \longrightarrow \mathbb{R}^m \rightsquigarrow d_x f : \mathbb{R}^n \longrightarrow \mathbb{R}^n$

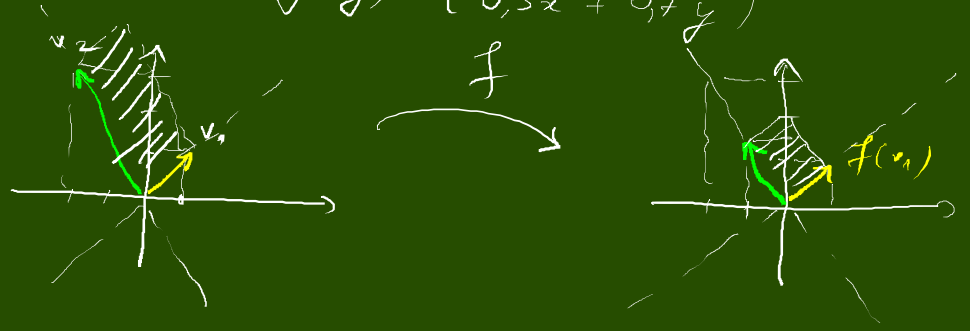
$x \in \mathbb{R}^n$ — application linéaire

$M(d_x f) = J_f(x)$

$f : V \longrightarrow V$ endomorphisme

Exemple $f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0{,}8x + 0{,}2y \\ 0{,}3x + 0{,}7y \end{pmatrix}$



$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $v_2 = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

$f(v_1) = v_1$

$f(v_2) = +\frac{1}{2} v_2$

$M(f)_B$

$B = \{e_1, e_2\}$

$M(f)_{B'}$

$B' = \{v_1, v_2\}$

$B = \{e_1, e_2\}$

$f(e_1) = \begin{pmatrix} 0,8 \\ 0,3 \end{pmatrix} = 0,8e_1 + 0,3e_2$

$f(e_2) = \begin{pmatrix} 0,2 \\ 0,7 \end{pmatrix} = 0,2e_1 + 0,7e_2$

$M(f)_B = \begin{pmatrix} 0,8 & 0,2 \\ 0,3 & 0,7 \end{pmatrix}$

$B' = \{v_1, v_2\}$

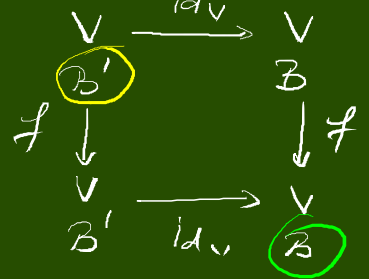
$f(v_1) = 1 \cdot v_1 + 0 \cdot v_2$

$f(v_2) = 0 \cdot v_1 + \frac{1}{2} v_2$

$M(f)_{B'} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$

$P_B^{B'} = (v_1 | v_2) = \begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix}$

→ base canonică !



Teoria :

$M(f)_B \cdot P$

$\begin{pmatrix} 0,8 & 0,2 \\ 0,3 & 0,7 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix} =$

$\begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$

$M(f)_{B'}$

$$f \circ id_V = f = id_V \circ f$$

$$M(f \circ id_V)_{B}^{B'} = M(id_V \circ f)_{B}^{B'}$$

$$M(f)_{B}^{B} \, M(id_V)_{B}^{B'} = M(id_V)_{B}^{B'} \cdot M(f)_{B'}^{B'}$$

$$M(f)_B \cdot P_B^{B'} = P_B^{B'} \cdot M(f)_{B'}$$

$$\boxed{P^{-1} \cdot M(f)_B \cdot P = M(f)_{B'}}$$

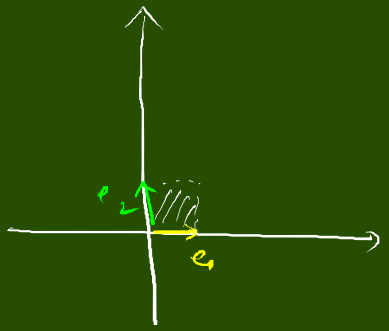
$A \in M_{m,n}(k)$ $\quad B = \{e_1 \dots e_n\}$ $\quad B' = \{e'_1 \dots e'_m\}$

$f : k^n \longrightarrow k^m$

$f(v) = Av$ $\quad (m,n)(n,1) = (m,1)$

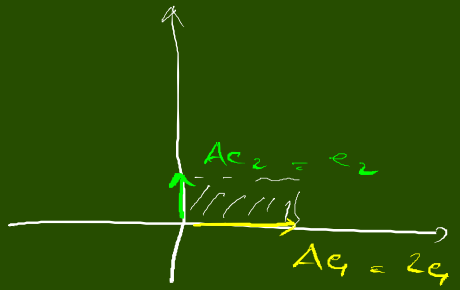
$\Rightarrow M(f)_{B'}^{B} = A$!

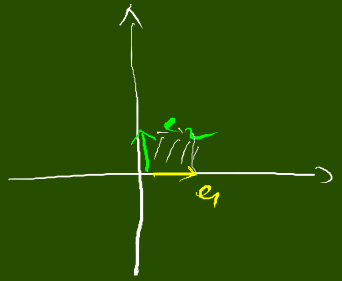
$A \in \mathcal{M}_2(\mathbb{R})$



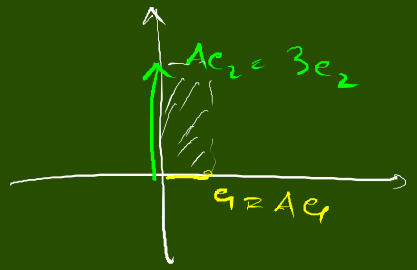
$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

$$v \longmapsto Av$$





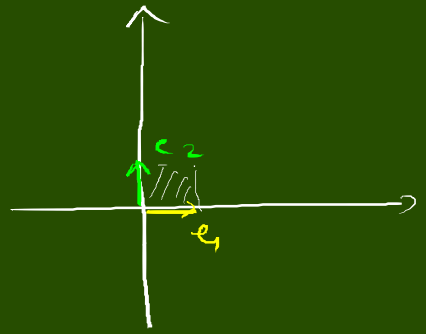
$$A = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$$



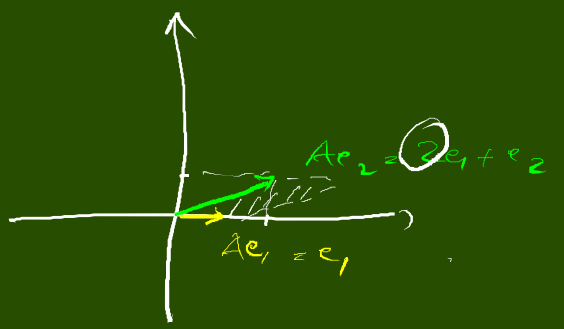
$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$Ae_2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2e_1 + e_2$$



A



forfecare / transvectie

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & c \end{pmatrix}$$ (shear)

tr. anamurfoce

AMES room

$$\begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}^n = \begin{pmatrix} \lambda_1^n & & 0 \\ & \ddots & \\ 0 & & \lambda_n^n \end{pmatrix}$$

$A = P D P^{-1}$ $\quad D = P^{-1} \cdot A \cdot P$

$k^n \xrightarrow{T(v) = Av} k^n$

$P = P_B^{B'}$ $\quad B$ $\quad B'$

B' k^n $\longrightarrow$ k^n B'

$M(f) = D$

$$\mathcal{J}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0{,}8x + 0{,}2y \\ 0{,}3x + 0{,}7y \end{pmatrix} = \underbrace{\begin{pmatrix} 0{,}8 & 0{,}2 \\ 0{,}3 & 0{,}7 \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \end{pmatrix}$$

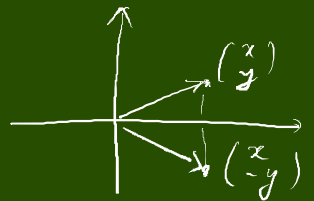
$\mathcal{J}(v_1) = Av_1 = v_1$ $\quad \lambda_1 = 1$

$\mathcal{J}(v_2) = Av_2 = \frac{1}{2} v_2$ $\quad \lambda_2 = \frac{1}{2}$

$A \cdot v = \lambda v$ → valoare proprie

→ $\neq 0$ vector propriu

$\mathcal{J} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ $\quad \mathcal{J}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \end{pmatrix}$



$y = 0$ $A\begin{pmatrix} x \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix}$

$\lambda_1 = 1$ $\quad v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$x = 0$ $A\begin{pmatrix} 0 \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -y \end{pmatrix}$

$\lambda_2 = -1$ $\quad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$A \in M_n(k)$ $\quad v \in k^n$ $\quad (v \neq 0_n)$ vector pr. eigenvector

$\lambda \in k$ $\quad$ val. pr. eigenvalue

$Av = \lambda v$

$Av - \lambda v = 0_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$

$(A - \lambda I_n) v = 0_n$ $\quad$ sistem omogen $n \times n$

are sol $\neq$ sol. banale $\iff$ $\det(A - \lambda I_n) = 0$

$n = 2$ $\quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$A - \lambda I_2 = \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix}$

$\det(A - \lambda I_2) = (a - \lambda)(d - \lambda) - bc =$

$$= \lambda^2 - \lambda \underbrace{(a+d)}_{\operatorname{tr}(A)} + \underbrace{(ad-bc)}_{\det(A)}$$

$A \in M_n(k)$

$\operatorname{grad} P_A = n$

$k = \mathbb{R}$

$P_A(x) \in \mathbb{R}[x]$

$\Big\downarrow$

$\mathbb{C}$

$$P_A(x) = x^2 - x \cdot \operatorname{tr}(A) + \det(A)$$

$\underline{n=2}$ $\quad A^2 - A \cdot \operatorname{tr}(A) + \det(A) \cdot I_2 = O_2$